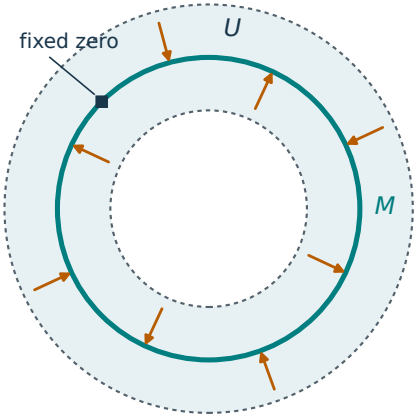
(a) Continuous limit retraction



$$P(x) = \lim_{k\to\infty} T^k x$$

$P : U \to M$ continuous, $\quad P|_M = \mathrm{id}_M$

$T = (I + \Pi_M)/2, \qquad P = \Pi_M$

(b) Admissible shapes

Singleton

Finite cross

branching

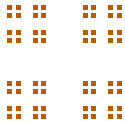
Filled square

corners + interior

Separate models:

no convexity or smoothness required

(c) Excluded accumulation



Perfect Cantor dust

$\{0\} \cup \{1/n : n \geq 1\}$

isolated points

Shrinking essential circles

Finite levels are drawn;

only the infinite limits are excluded.

Local RLEB + continuous single-valued full proximal map T + uniform vanishing physical length tails.